ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M037 **Name:** Kunal Thakur

Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) LSB = $5\text{ V}/2^{12} = 5/4096 = 1.2207\text{ mV} = 1.2207\times10^{-3}$ V.

ΔF = LSB/S with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 = \mathbf{1.22\times10^{-4}\ N}$.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain G/(1 − F) = 0.090635/0.18127 = 0.5 ✓.

$$\mathbf{x_{k+1} = 0.8187\,x_k + 0.0906\,u_k}$$

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally because the derivative of a sampled signal is not defined.

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s$ = 0.001 s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a)\,z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a)\,z^{-1}]$.

Dividing through by (1 + a): $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: **y_k = 0.05912 (u_k + u_{k-1}) + 0.88177 y_{k-1}**. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): y = 2(0.05912)u + 0.88177 y ⇒ y(1 − 0.88177) = 0.11824 u ⇒ y/u = 0.11824/0.11823 = 1.000. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: O = [C; CA] = [1, 0; 0, 1], full rank, so the observer poles can be placed arbitrarily.

With L = $[\ell_1, \ell_2]^T$ and C = [1, 0], LC = $[\ell_1, 0; \ell_2, 0]$, so A − LC = $[-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: det(sI − (A − LC)) = det$[s + \ell_1, -1; \ell_2, s + 2]$ = $(s + \ell_1)(s + 2) - (-1)(\ell_2)$ = $s^2 + (2 + \ell_1)\, s + (2\ell_1 + \ell_2)$.

Both estimator poles at s = −10 means the desired polynomal is $(s + 10)^2 = s^2 + 20\, s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

L = [18, 64]^T. Verification: A − LC = [−18, 1; −64, −2] has trace −20 and determinant (−18)(−2) − (1)(−64) = 36 + 64 = 100, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Observer advantage: because it is a copy of the plant corrected by the encoder error, it gives a velocity estimate without the delay that a filter needs for noise rejection, and it can also estimate quantities no filter on θ can, such as an unknown load torque.

A filter is the safer choice when the model is wrong or the true input is not known, e.g. unmodelled friction or backlash, a changing payload, or actuator saturation: the observer would then be driven by a wrong u and drift, while the filter only uses measured data.

Note: observer poles are usually placed several times faster than the controller poles.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\, u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\, r - k_1\theta - k_2\dot{\theta}$.

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$, i.e. $s^2 + 4s + 8 = 0$ so the poles are s = −2 ± 2j.

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) $G_{DC} = bN_{ff}/(c + bk_1) = 6/7.5 = 0.8$, so $\theta_{ss} =$ **0.8**.

Percentage error = 0.8 × 100 = **80%**.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\, u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With r = 0 (r only shifts the equilibrium) the closed-loop matrix is A_{cl} = [0, 1, 0; $-(c + bk_1)$, $-(a + bk_2)$, bK_I; −1, 0, 0].

Characteristic polynomial det(sI − A_{cl}), expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired poles −4 and −2 ± 2j: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by (s + 4): $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with a = 2, b = 1, c = 6: s^2: $2 + k_2 = 8 \Rightarrow$ **k_2 = 6**. s^1: $6 + k_1 = 24 \Rightarrow$ **k_1 = 18**. s^0: $b\, K_I = 32 \Rightarrow$ **K_I = 32**.

Check: all coefficients of the desired polynomal are positive and (8)(24) = 192 > 32, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Because the closed loop is stabel, the steady-state error is zero for any b and c.

Failure mode: integrator windup. Remedy: clamp the integrater output (anti-windup).